

Bridge Building Competition

Event Description

The Bridge Building Competition is a practical engineering challenge designed to test participants' creativity, structural understanding, problem-solving skills, and ability to design an efficient bridge using limited materials. Participants will construct a model bridge capable of spanning the required gap and carrying a specified load while maintaining stability. The competition focuses on applying basic principles of structural engineering in a simple and practical manner.

Participants are encouraged to develop lightweight, stable, strong, and innovative bridge designs while making efficient use of the available materials. The event provides an opportunity for students from different engineering disciplines to demonstrate their technical knowledge, teamwork, creativity, and practical engineering skills.

Eligibility Criteria

- The competition is open to students from Civil, CSE, Mechanical, and other eligible engineering as well as non- tech related Departments.
- Each team must consist of a minimum of 2 students and a maximum of 5 students.
- A team cannot have more than 5 members.
- Each participant may be a member of only one team.
- All team members must be registered participants of the event.
- Teams must construct and present their bridge according to the given competition requirements and rules.

Evaluation / Scoring Criteria

1. Span Requirement

- The bridge deck must span a minimum clear gap of 40 cm.
- The bridge must have a minimum 10 cm bearing/support length on each side.
- Therefore, the minimum overall length of the bridge should be approximately 60 cm.
- The bridge must be properly positioned on the designated supports before testing.

2. Stability

- The completed bridge must remain freestanding without any external support for a minimum of 15 seconds.
- The bridge should remain stable and in its intended position during this period.
- Any significant collapse, slipping, or loss of structural stability will result in failure of the stability requirement.

3. Load-Bearing Capacity

- Loads will be applied gradually at the mid-span of the bridge.
- The load will be increased step-by-step until the bridge reaches its failure point.
- The maximum load successfully carried by the bridge will be considered for scoring.
- Higher load-bearing capacity will result in a higher score.

4. Efficiency (Strength-to-Material Ratio)

- Teams will be evaluated on how efficiently they use the available materials.
- All materials will be provided by the organising committee.

5. Design & Innovation

- The bridge will also be evaluated based on its design, creativity, workmanship, and effective use of materials.
- Innovative structural arrangements and practical engineering solutions will be given additional consideration.
- The design should demonstrate a clear balance between strength, stability, and material efficiency.

6. Assessment & Scoring

Evaluation Criterion| Maximum Points

Load-Bearing Capacity| 10

Stability| 10

Strength-to-Material Ratio| 10

Design & Innovation| 10

Total Score| 40 Points

Certificates:

All participants will be awarded Certificates of Participation. Winners will also receive Winner Certificates along with their respective prizes.

Important Notes

- The bridge must satisfy the minimum 40 cm clear span requirement to qualify for final scoring.
- All bridges will be tested under the same conditions to ensure fairness.
- The load will be applied carefully and gradually at the mid-span.
- The decision of the evaluation/judging panel regarding structural failure and final scoring will be considered final.
- Materials will be provided by the committee, and no outside materials will be allowed.